RAPT-CLIP: Unified Framework for Context-Aware Video Emotion Recognition

Detailed Micro-Architectural Topology with Explicit Intermediate Stages between CMAF and Temporal Transformer

VISUAL BRANCH: Multi-Stream Extraction, Frame-Level CMAF Fusion, Temporal Bridging & Unified Modeling

Face Stream

$F_{1..T}$ Crops

(B, T, 3, 224, 224)

CLIP ViT-B/16

12 Layers, 196 Patches

$\mathbf{z}_t^{face} \in \mathbb{R}^{B \cdot T \times 512}$

Shared

Body Stream

$B_{1..T}$ Crops

(B, T, 3, 224, 224)

CLIP ViT-B/16

Shared Weights θ_{ViT}

$\mathbf{z}_t^{body} \in \mathbb{R}^{B \cdot T \times 512}$

Shared

Context Stream

$C_{1..T}$ Scene

(B, T, 3, 224, 224)

CLIP ViT-B/16

Shared Weights θ_{ViT}

$\mathbf{z}_t^{ctx} \in \mathbb{R}^{B \cdot T \times 512}$

Face Adapter (EAA)

Bottleneck $512 \to 128 \to 512$
Residual + ReLU

$\mathbf{f}_t^{face} \in \mathbb{R}^{B \cdot T \times 512}$

Mod. Drop (p=0.3)

CMAF Cross-Attention Per-Frame Core

Tridirectional Cross-Attn ($h = 4$):
- Q_F x [K_B, K_C] -> f'_F
- Q_B x [K_F, K_C] -> f'_B
- Q_C x [K_F, K_B] -> f'_C

3 Residual LayerNorm Outputs (each 512-d)

$\mathbf{f}_t'^F, \mathbf{f}_t'^B, \mathbf{f}_t'^C$

3x512

Stage 1: Channel Concatenation

$\mathbf{f}_t = [\mathbf{f}_t'^F \| \mathbf{f}_t'^B \| \mathbf{f}_t'^C]$
torch.cat(dim=-1)
$512 \times 3 \to 1536$

$\mathbf{f}_t \in \mathbb{R}^{(B \cdot T) \times 1536}$

$\mathbf{f}_t$

Stage 2: Temporal Sequence Reshape

view($B, T = 16, 1536$)
Frames -> Video Series
$[\mathbf{f}_1, \mathbf{f}_2, \ldots, \mathbf{f}_{16}]$

$\mathbf{F}_{seq} \in \mathbb{R}^{B \times 16 \times 1536}$

Temporal Pos. Embedding

$\mathbf{E}_{pos}$ Learnable

(1, 16, 1536)

$\oplus$

X

Unified Temporal Transformer

1 Layer, $T = 16$, $h = 8$
Pre-LN + MHSA(8)
+ Residual FFN(1024)

$\mathbf{H} \in \mathbb{R}^{B \times 16 \times 1536}$

$\mathbf{h}_t$

Attention Pooling

$\mathbf{u}_t = \tanh(\mathbf{W}_1 \mathbf{h}_t)$
$\alpha_t = \text{softmax}(\mathbf{w}_2^\top \mathbf{u}_t)$
$\mathbf{F} = \sum \alpha_t \mathbf{h}_t$

(B, 1536)

F

v

$\mathbf{W}_p$
ℓ_2

512

Cosine Sim: $\mathbf{v}^\top \mathbf{t}_k$

TEXT BRANCH: Dual-Prompt Learning (CoOp Tuning + Hand-Crafted Clinical Anchors)

Prompt Ensemble (View 1)

K=5 classes x P=3 templates
e.g., 'a photo of [CLS] student'

(K \cdot P, \text{tokens})

Prompt Learner (CoOp)

M=8 Learnable Soft Tokens
$[SOS] \oplus [V]_1 \ldots [V]_M \oplus [CLS]$

(K \cdot P, 77, 512)

Semantic Clinical Descriptors (View 2 - Anchors)

Detailed physiological cues: e.g., 'droopy eyelids, head tilting down, yawning' (Fatigue)

(K, 77, \text{tokens})

CLIP Text Encoder (12-Layer Transformer)

Shared Embedding Matrix
EOT Token Extraction
Text Projection $\to \ell_2$-norm

out: (K, 512)

View 1: $\mathbf{T}^{learn}$

Ensemble Averaged Learned Embeddings

(K, 512)

View 2: $\mathbf{T}^{desc}$

Frozen Semantic Anchor Embeddings

(K, 512)

Text Prototypes $\mathbf{t}_k$

Semantic Regularization (MI Loss + DC Loss)

- L_MI: Symmetrical InfoNCE
  Aligns T^learn <-> T^desc
- L_DC: Decorrelation Penalty
  Minimizes sum_{i!=j} (t_i^T t_j)^2

L_reg = lambda_MI * L_MI + lambda_DC * L_DC

LDAM Classification Loss (Imbalance-Aware)

Logits: z_k = (v^T t_k) / tau
Margin: Delta_y = C / N_y^(1/4)
scale s=30, temp tau=0.07
Balances rare vs majority classes

L_total = L_LDAM + 0.1*L_MI + 0.1*L_DC

DETAILED INTERNAL MICRO-ARCHITECTURES (Bridging & Sub-Module Breakdowns)

(a) EAA Bottleneck Adapter

- Input: $\mathbf{z}_t^{face} \in \mathbb{R}^{512}$ (CLIP Token)
- Linear Down: $512 \to 128$ (dim // 4)
- Activation: ReLU + Dropout(0.5)
- Linear Up: $128 \to 512$
- Residual Add: $\mathbf{z}_t^{face} + \Delta\mathbf{z}_t^{face}$
- Modality Dropout: $p = 0.3$ (Occlusion-proof)

Preserves base CLIP semantic space

(b) CMAF-to-Temporal Bridging Flow

1. Cross-Attn: Q_F, Q_B, Q_C attend pairs -> f'_F, f'_B, f'_C
2. Channel Concat: [f'_F || f'_B || f'_C] -> f_t in R^(B*T x 1536)
3. Temporal Stacking: contiguous().view(B, T=16, 1536)
   Groups flat per-frame vectors into video sequence F_seq
4. Positional Add: X = F_seq + E_pos (1 x 16 x 1536)
   Breaks permutation-invariance for chronological attention

Essential bridge from frame-level to video-level

(c) Dynamic Attention Pooling

- Input Tokens: $\mathbf{H} = [\mathbf{h}_1, \ldots, \mathbf{h}_{16}] \in \mathbb{R}^{16 \times 1536}$
- Scoring MLP: $\mathbf{u}_t = \tanh(\mathbf{W}_1 \mathbf{h}_t + \mathbf{b}_1) \in \mathbb{R}^{512}$
- Softmax Weights: $\alpha_t = \frac{\exp(\mathbf{w}_2^\top \mathbf{u}_t)}{\sum_i \exp(\mathbf{w}_2^\top \mathbf{u}_i)}$
- Aggregated Vector: $\mathbf{F} = \sum_{t=1}^{16} \alpha_t \mathbf{h}_t$
- Linear Head: $\mathbf{W}_{proj}(1536 \to 512) + \ell_2$-norm

Pinpoints dynamic peak emotional keyframes

(d) Multi-Objective Loss Joint

- L_LDAM: Tail-margin penalty (s=30, tau=0.07)
- L_MI: Symmetrical InfoNCE (T=0.07)
- L_DC: Text feature orthogonalization
- L_total = L_LDAM + 0.1*L_MI + 0.1*L_DC
- Canonical SOTA: 77.06% UAR / 77.65% WAR

Guarantees high UAR without class collapse